

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 212 — Past Questions & Materials
by Renance

Course: MTH 212
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

This study pack was compiled and restructured by Renance for MTH 212 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — Algebra Set and Logic 2020 2021 First Semester MTH 212

Level 200 • First Semester • PastQuestion

MINATION 2020/2021 SESSION

E& CODE: ALGEBRA, SET AND LOGIC (MTH 212)

Time: 2 hours

“lined on the

set R of real numbers by $a * b = a + b^4$ — for all $a, b \in R$

4) Is the Operation $*$ Commutative (3 marks)

b) Is the Operation $*$ Associative (6.5 marks)

€) find the identity element e Of the set and Inverse element a^{-1} of a . (8 marks)

Question 2 (17,5 marks)

a) Test the Validity of the following argument

$p \rightarrow q$

1° TTT study. then y

P2: If 1 do not pl

P3: But y failed

Vill not fail Mathematics

ay basketball. then y w

ill study

Mathematics

NE CRW OR EW Edie separa

b) Prove the

Question 3 (17.

a) Fj

following

Mon divisor (2cd) of the following

1) 1236 4323 and the Value of the integers x and y such that $\gcd = 4323x + 1236y$. (6 marks)

(i) 12367 24983 and the Value of the integers x and y such that $\gcd = 24983x + 12367y$. (6 marks)

b) Prove that if $a|b$ and $b|c$. then $a|c$ and if $d|x$ and $d|y$ then for any integers a, b we have $d|(ax + by)$. (5.5 marks)

Question 4 (17.5 marks)

4) Define a group and prove that identity element of a group

b) Show that $(G,$

$\cdot$) and the inverse of an element are unique (8.5 marks)

$(G, *)$ is non-abelian group, where G

$= \{(a, b) : a, b \in \mathbb{R} \setminus \{0\}\}$ and $*$ is defined on G by $(a, b) * (c, d) =$

(ad, cd) (9 marks)

(ac, bce +

Question 5 (17,

> marks)

a) (i) Given $A = \{1,2\}$, $B = \{x,y,z\}$ and $C = \{3,4\}$. Find $A \times B \times C$

(i) Let $A = \{1,2,3\}$, $B = \{a,b,c\}$ and $C = \{X, y, 2\}$. Consider the following relations R and S from A to B and from B to C

respectively, $R = \{(1, b), (2, Q), (2, c)\}$; $S = \{(a, y), (b, x), (c, y), (c, 2)\}$. Find the composition relation $R \circ S$ and the product of

the matrices M_R and M_S . (8.5 marks)

b)

Consider the following five relations on the set $A = \{1, 2, 3\}$

$R = \emptyset$ empty relation; $A \times A$ Universal relation

$S = \{(1,1), (1,2), (2,2), (2,3)\}$; $T = \{(1,2), (2,3)\}$

Question 6 (17.5 marks)

1. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by

$f(x) = \begin{cases} x & \text{if } x \text{ is even} \\ x+1 & \text{if } x \text{ is odd} \end{cases}$ where x is a real number. (2 marks)

4) (i) State the largest domain of each of the functions $f(x) = 72$, and $g(x) = \frac{1}{x}$ Where x is a real number

a) 7. (ii) Find the inverse function g^{-1} of g and the composite function $g \circ f(x)$. In each case, state the domain. (5 marks)

li) Find the inverse function g^{-1} of g and the composite function $g \circ f(x)$. In each case, state the domain. (5 marks)

ij) Find the domain of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$f(x) = \begin{cases} x & \text{if } x \leq 10 \\ x+1 & \text{if } x > 10 \end{cases}$ Alt he 50 science students in a school were asked their subject combination. 18 of the students offered

Mathematics and Biology, 9 students offered Chemistry and Biology while 5 students offered the three subject combinations.

offered Chemistry while 16 offered Biology. 7 students offered Further Mathematics and Biology, 9 students offered Chemistry and Biology while 5 students offered the three subject combinations.

Mathematics, 2]

Mathematics and Chemistry, 8 students offered Further

Mathematics and Chemistry, 8 students offered Further

Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.

Mathematics and Chemistry, 8 students offered Further Mathematics and Chemistry, 8 students offered Further Mathematics.